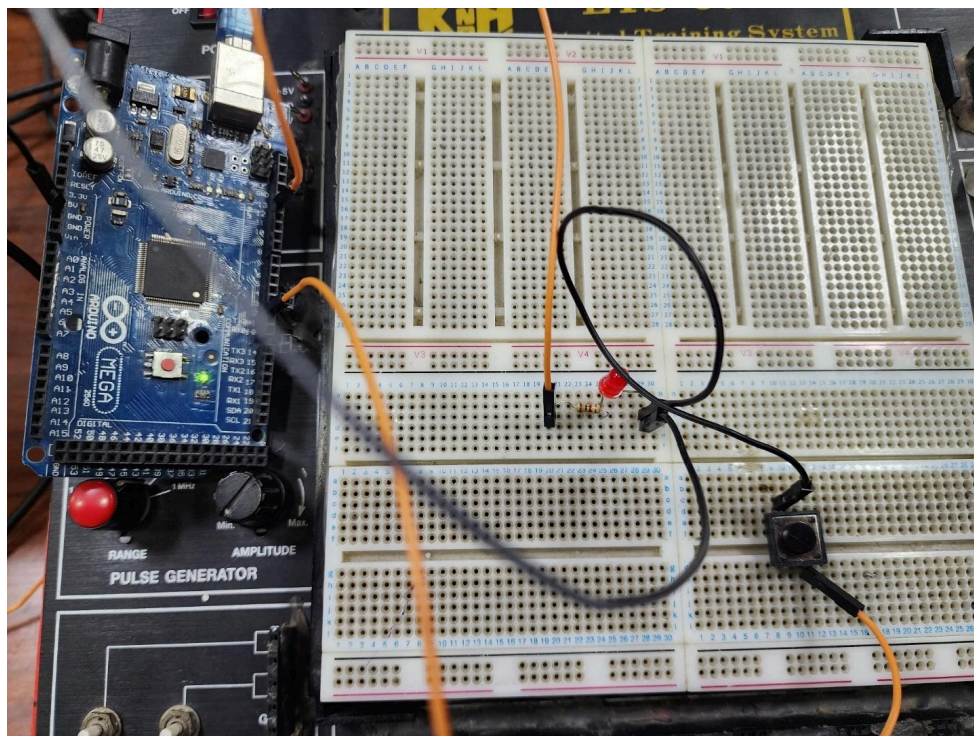


Lab #2

Question #1:

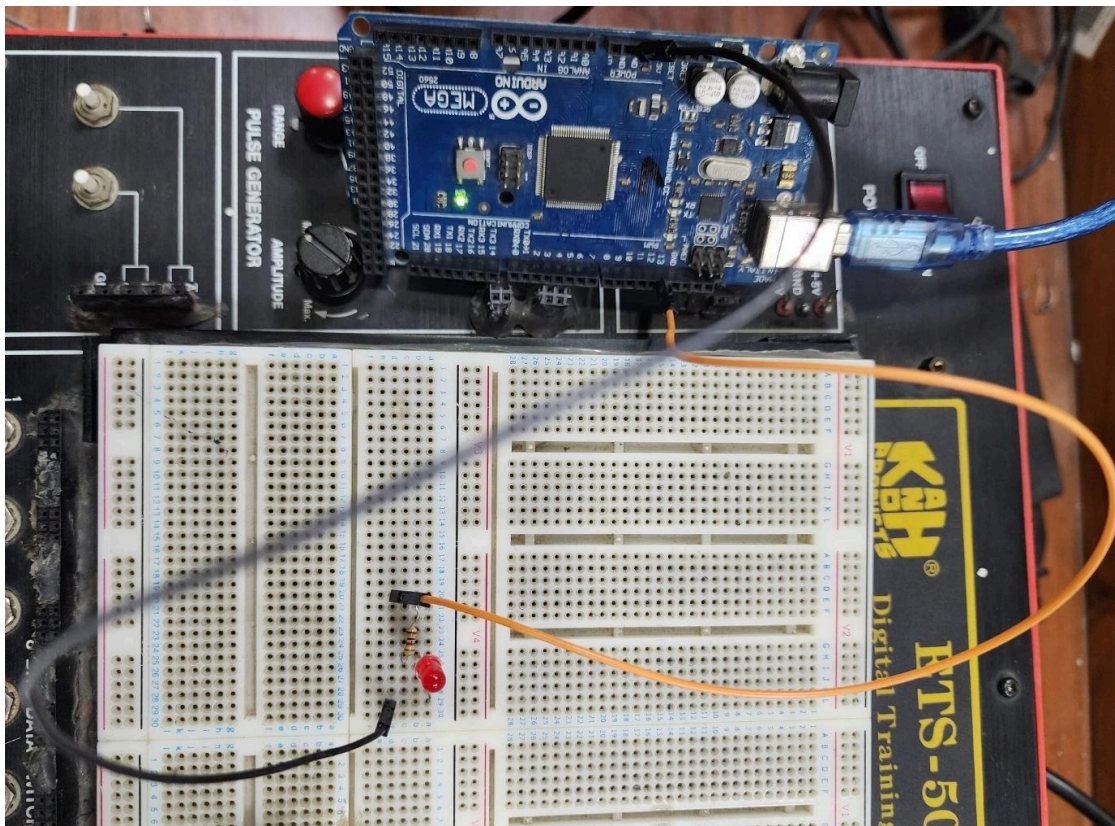
```
sketch_jan29a.ino

1  const int buttonPin = 2;
2  const int ledPin = 13;
3
4  void setup() {
5
6      pinMode(ledPin, OUTPUT);
7      pinMode(buttonPin, INPUT_PULLUP);
8  }
9
10 void loop() {
11     int buttonState = digitalRead(buttonPin);
12
13     if (buttonState == LOW) {
14         digitalWrite(ledPin, HIGH); // LED ON
15     } else {
16         digitalWrite(ledPin, LOW); // LED OFF
17     }
18 }
19
20
```



Question #2:

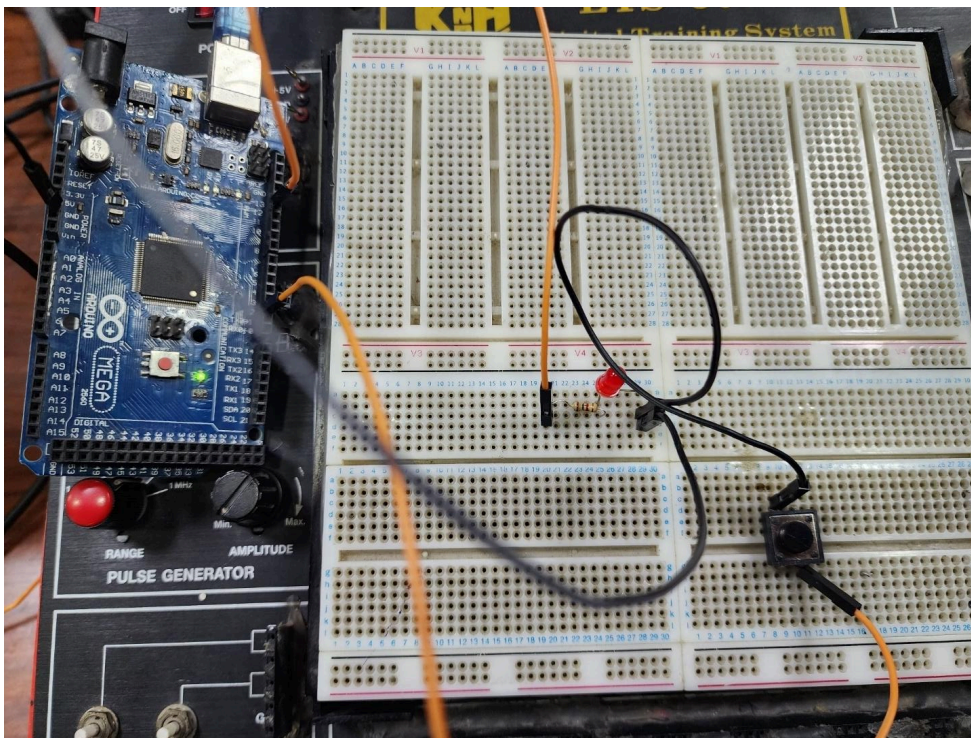
```
1  int number = 45;
2  const int ledPin = 13;
3
4  void setup() {
5      pinMode(ledPin, OUTPUT);
6
7      if (number > 50) {
8          digitalWrite(ledPin, HIGH); // LED ON
9      } else {
10         digitalWrite(ledPin, LOW); // LED OFF
11     }
12 }
13
14 void loop() {
15 }
```



Question #3:

sketch_jan29a.ino

```
1
2  const int buttonPin = 7;
3  const int ledPin = 13;
4
5  void setup() {
6    pinMode(ledPin, OUTPUT);
7    pinMode(buttonPin, INPUT_PULLUP);
8  }
9
10 void loop() {
11   if (digitalRead(buttonPin) == LOW) {
12     digitalWrite(ledPin, HIGH); // LED ON
13   } else {
14     digitalWrite(ledPin, LOW); // LED OFF
15   }
16 }
```



4) Difference Between Flash Memory and SRAM

Feature	Flash Memory	SRAM
Type	Non-volatile	Volatile
Data Retention	Keeps data after power OFF	Data lost when power OFF
Used For	Storing program/code	Storing temporary variables
Size	Larger	Smaller
Speed	Slower	Faster

Daily life items using Flash Memory:

- USB Flash drive
- SSD (Solid State Drive)
- Memory card (SD card)
- Smartphone internal storage
- BIOS/UEFI chip
- Digital cameras

5) Advantage of EEPROM

Advantages:

- Non-volatile (data saved after power OFF)
- Can store small important data
- Rewritable many times

When to use EEPROM?

Storing user settings

Saving calibration values

Storing WiFi credentials in IoT

Saving last system state

6) Disadvantages of uploading code using Arduino Bootloader

Disadvantages:

- Takes extra memory space (bootloader uses Flash)
- Slight delay at startup
- Slower uploading compared to programmer

How to avoid it?

- Use External Programmer (ISP programmer)
- Burn code directly using USBasp
- Remove bootloader and upload via ICSP

7) Why is ADC used for analog pins in microcontrollers?

Microcontrollers work with digital values (0 and 1).

But sensors (temperature, LDR, potentiometer) give analog signals (continuous voltage).

ADC (Analog to Digital Converter) converts:

Analog voltage → Digital value

Example:

0V → 0

5V → 1023 (for 10-bit ADC in Arduino)

So ADC allows microcontrollers to read real-world analog signals.

8) Difference Between Arduino and Raspberry Pi

Feature	Arduino	Raspberry Pi
Type	Microcontroller	s Microprocessor (Mini computer)
OS	No OS	Runs Linux OS
Speed	Slower	Much Faster
Multitasking	No	Yes
Power Consumption	Low	Higher
Best For	Real-time control, sensors AI, Web server, Image processing	
Programming	C/C++ Python, C++, etc	

Simple Understanding:

Arduino → Control devices (LED, motors, sensors)

Raspberry Pi → Full computer (AI, IoT server, camera processing)